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Palladium-Catalyzed Triple Cyclization of 2,7-Alkadiynylic Carbonates with 2-Butyne-1,4-diol or 2-Butyne-1,4-disulfonamides

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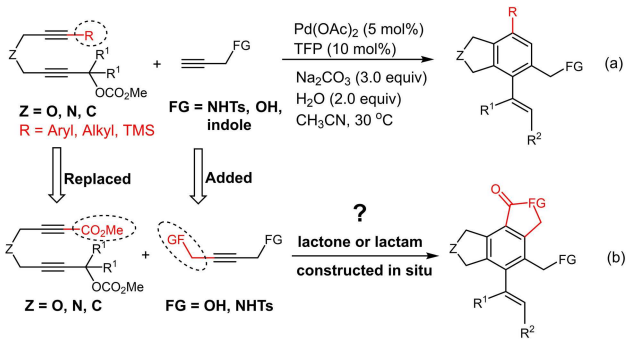
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Abstract: A palladium-catalyzed highly *E*-selective triple cyclization of 2,7-alkadiynylic carbonates with 2-butyne-1,4-diol or 2-butyne-1,4-disulfonamides to construct different tricycles such as 6,8-dihydrobenzo[1,2-c:3,4-c′]difuran-1(3*H*)-one and 1,3,6,7-tetrahydro-8*H*-furo[3,4-e]isoindol-8-one derivatives has been reported. The alkene group produced and the nucleophilic unit kept in the product provide further opportunity to construct more complicated polycycles. The control experiments showed that the reaction mainly go through an intermolecular insertion of the alkynes before the lactonization or lactamization.

Keywords: palladium; benzopolycycle; 2,7-alkadiynylic carbonates; 2-butyne-1,4-diol; 2-butyne-1,4-disulfonamide

Polycycles containing heterocycles such as lactones and lactams are of great importance due to their potential biological activities.[1] Thus, they have drawn much attention of the chemists to develop more efficient synthetic methods.[2–8] Common approaches to construct these types of skeletons include: (1) intramolecular [2+2+2] cycloaddition of 1,6,11-triynes;[3,4] (2) the Hexadehydro-Diels-Alder (HDDA) reaction of 1,3,8-triynes followed by the addition with intramolecular or intermolecular electrophiles;[5,6] (3) intra- or intermolecular domino reactions consisting of oxidative addition, carbopalladation, Heck couplings and consecutive 6π-electrocyclizations of alkenyl halides with one or more alkyne moieties;[7] (4) intermolecular tandem reactions consisting of oxidative addition, Sonogashira couplings and sequenced Garratt-Braverman cyclization of alkenyl iodides with 1,6-diynes.[8] Nowadays, palladium-catalyzed cyclization of propargyl alcohol derivatives with unsaturated hydrocarbons such as alkenes, alkynes, and allenes has been a powerful tool to prepare polycycles.[9] Recently, our group developed an efficient benzofuran construction via a Pd-catalyzed intermolecular annulation of 2,7-alkadiynylic carbonates with functionalized terminal alkynes (Scheme 1a).[10] We assumed that if the R group at 8-position is replaced with an ester group, the newly designed 2,7-alkadiynylic carbonates may react with symmetric internal alkynes such as 2-butyne-1,4-diol or 2-butyne-1,4-disulfonamides to form tricyclic compounds with an extra lactone or lactam unit (Scheme 1b).

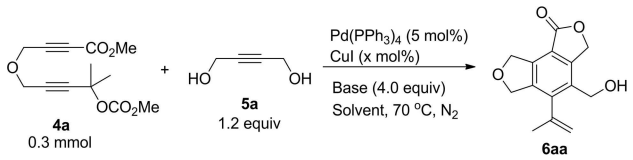
With this notion in mind, we initially conducted the reaction of 2,7-alkadiynylic carbonate (**4a**) with 2-butyne-1,4-diol (**5a**) under the catalysis of $Pd(PPh_3)_4$ (5 mol%) and Et_3N (4.0 equiv.) at 70 °C in THF. To our delight, the expected tricyclic product **6aa** was observed in 33% NMR yield after 7 hours (entry 1, Table 1). Interestingly, the reaction became more effective in the presence of a small quantity of copper



Scheme 1. The concept of this work.

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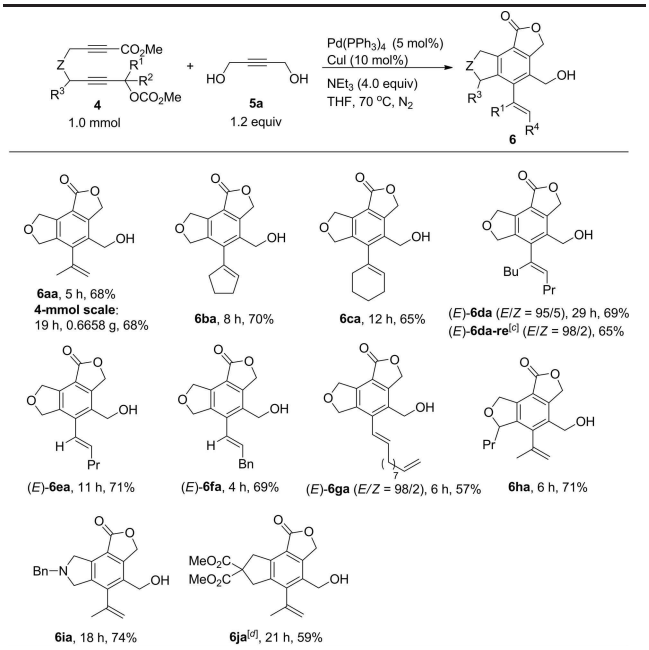
Table 1. Optimization of the conditions.[a]



Entry	CuI (x mol %)	Base	Solvent	Time (h)	Yield of **6aa** (%)[b]
1	0	Et_3N	THF	7	33
2	5	Et_3N	THF	12.5	69
3	**10**	**Et_3N**	**THF**	**3**	**74**
4	20	Et_3N	THF	2	73
5	40	Et_3N	THF	3	62
6	10	Na_2CO_3	THF	2	64
7	10	Cs_2CO_3	THF	3.5	-
8	10	NaOMe	THF	16	31
9	10	DBU	THF	2	-
10	10	Et_3N	DMSO	4	53
11	10	Et_3N	DMF	2	58
12	10	Et_3N	DCE	2	64
13[c]	10	Et_3N	THF	3.5	64
14[d]	10	Et_3N	THF	14	64

[a] Reaction conditions: **4a** (0.3 mmol), **5a** (1.2 equiv.), Pd(PPh_3)$_4$ (5 mol%), CuI (x mol%), base (4.0 equiv.) in solvent (3.0 mL) at 70 °C unless otherwise noted.
[b] Determined by ^{1}H NMR analysis of the crude product.
[c] $Pd(OAc)_2$ (5 mol%) and TFP (10 mol%) were used instead of Pd(PPh_3)$_4$.
[d] $Pd(OAc)_2$ (5 mol%) and PPh_3 (10 mol%) were used instead of Pd(PPh_3)$_4$.

Table 2. The reactions of 8-methyoxycarbonyl substituted 2,7-alkadiynylic carbonates with 2-butyne-1,4-diol.[a,b]



[a] Reaction conditions: **4** (1.0 mmol), **5a** (1.2 equiv.), Pd(PPh_3)$_4$ (5 mol%), CuI (10 mol%), NEt_3 (4.0 equiv.) in THF (10 mL) at 70 °C unless otherwise noted.
[b] Isolated yield.
[c] After recrystallization.
[d] Reaction conditions: **4j** (1.0 mmol), **5a** (1.2 equiv.), Pd(OAc)$_2$ (5 mol%), TFP (10 mol%), CuI (10 mol%), Na_2CO_3 (3 equiv.), H_2O (2.1 equiv.), in CH_3CN (10 mL) at 70 °C.

iodide (entries 2~5, Table 1). When CuI (10 mol%) was added, the yield of **6aa** reached the highest (entry 3, Table 1). Various bases such as Na_2CO_3, Cs_2CO_3, NaOMe and DBU were subsequently investigated, among which no one was better than NEt_3 (entries 6–9, Table 1). Further solvent screening showed that the reactions in DMSO, DMF, DCE delivered poorer results than that in THF (entries 10–12, Table 1). When Pd(PPh_3)$_4$ was replaced with Pd(OAc)$_2$/TFP (TFP = tri(2-furyl)phosphine) or Pd(OAc)$_2$/PPh_3, the reactions afforded **6aa** in lower yields (entries 13–14). Thus, the optimal conditions were defined as follows: Pd(PPh_3)$_4$ (5 mol%), CuI (10 mol%), and NEt_3 (4.0 equiv.) in THF at 70 °C.

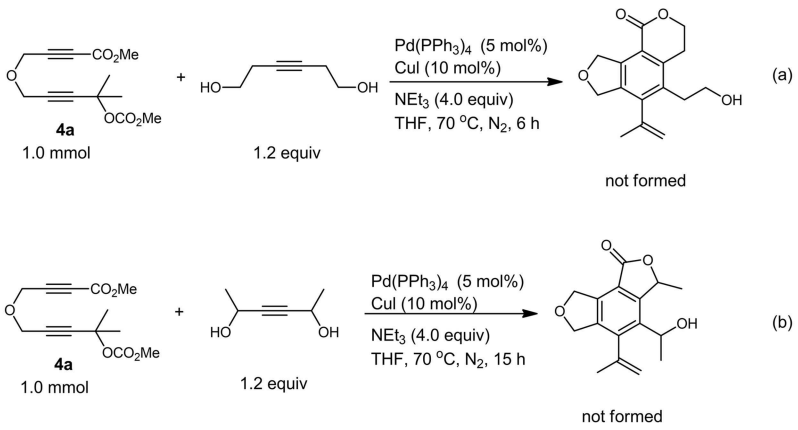
With the optimal conditions in hand, we first examined the reactivity of various 8-methyoxycarbonyl substituted 2,7-alkadiynylic carbonates with 2-butyne-1,4-diol (Table 2). R^1, R^2 could be the same alkyl group, such as methyl (**6aa**), cyclopentyl (**6ba**), and cyclohexyl (**6ca**). The corresponding tricyclic lactones were isolated in 65~70% yields. The reaction of **4d**, in which R^1, R^2 were both butyl groups, afforded (*E*)-**6da** in 69% yield with an *E*/*Z* ratio of 95/5. After recrystallization, 65% yield of (*E*)-**6da** was obtained with an improved *E*/*Z* ratio of 98/2. Secondary 2,7-alkadiynylic carbonates were also applicable and transferred into the tricyclic lactones (*E*)-**6ea**~(*E*)-**6ga** in 63~71% yields with excellent *E*/*Z* ratios. In addition, a propyl group could be introduced into 4-position of the starting material (**6ha**). Such a strategy was also compatible for the construction of nitrogen- or carbon-tethered tricyclic lactones (**6ia**–**6ja**). It's worth mentioning that the preparation of **6aa** could be easily scaled up to a 4-mmol scale.

However, when we extended the carbon chain on each side by one carbon atom, i.e., hex-3-yn-1,6-diol (eq. a in Scheme 2) or used hex-3-yn-2,5-diol (eq. b in Scheme 2), the reaction was complicated and the expected product was formed.

Furthermore, it was exciting to find that 2-butyne-1,4-diol could be replaced with 2-butyne-1,4-disulfonamides. The corresponding tricyclic lactams were delivered in moderate yields of 37~58% (Table 3). When 2-butyne-1,4-diol was replaced with 2-butyne-1,4-disulfonamides, the products were impure after chromatography on silica gel, which should be further purified by recrystallization (DCM/hexane) to afford the pure products. We reasoned that this is the reason

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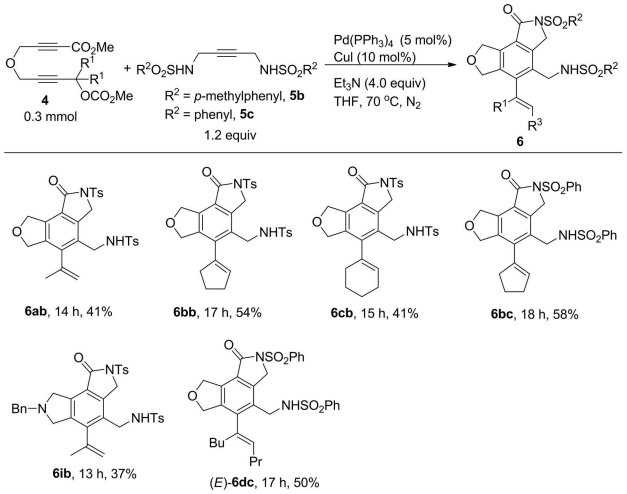


Scheme 2. The reactions of 2,7-alkadiynylic carbonate with hex-3-yn-1,6-diol or hex-3-yne-2,5-diol.

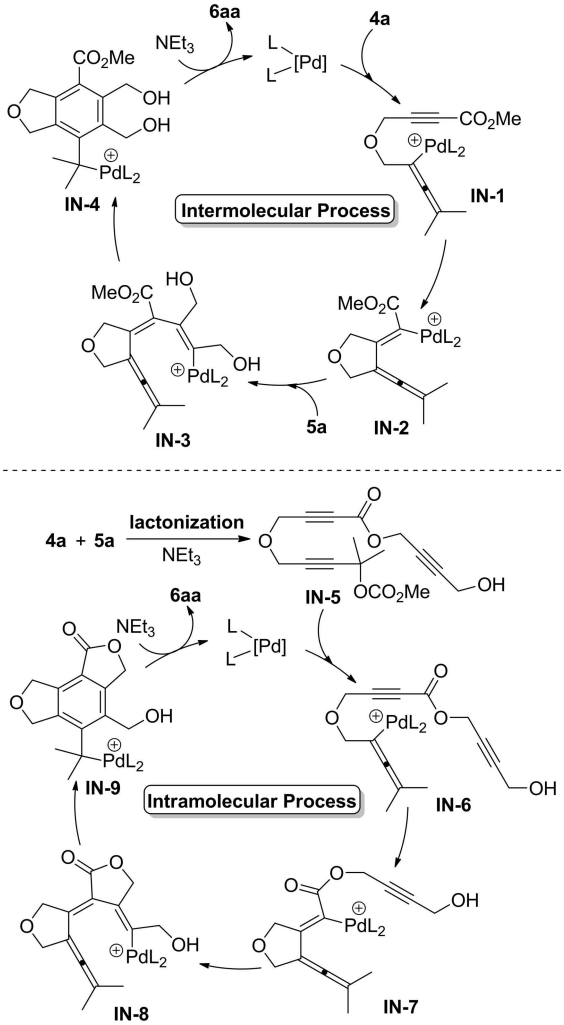
for the low isolated yield in comparison to 2-butyne-1,4-diol. The structure of the target tricycle was unambiguously established by the X-Ray single crystal diffraction analysis of **6bb**·$CHCl_3$ (Figure 1).[11]

On the basis of our experimental results, we proposed two possible mechanism routes as shown in Scheme 3 by taking the reaction of **4a** and **5a** as an example. For the **intermolecular process**, oxidative addition of **4a** with Pd(0) would afford the allenylpalladium **IN-1**,[12] which undergoes intramolecular insertion of the triple bond to generate the alkenylpalladium species **IN-2**.[9j–m,10] The species **IN-2** would undergo intermolecular *syn*-carbopalladation of the C≡C bond of **5a** to produce a new alkenylpalladium intermediate **IN-3**.[10] Then intramolecular carbopalladation of allene would form a new benzylpalladium intermediate **IN-4**, which finally form the tricyclic

Table 3. The reactions of 8-methyoxycarbonyl substituted 2,7-alkadiynylic carbonates with 2-butyne-1,4-disulfonamides.[a,b]



[a] Reaction conditions: **4** (0.3 mmol), **5** (1.2 equiv.), $Pd(PPh_3)_4$ (5 mol%), CuI (10 mol%), NEt_3 (4.0 equiv.) in THF (6 mL) at 70 °C unless otherwise noted.
[b] Isolated yield.



Scheme 3. Proposed mechanism: Intermolecular Process vs. Intramolecular Process.

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product **6aa** after the β-H elimination and lactonization in the presence of NEt_3. For the **intramolecular process**, the lactonization of **4a** and **5a** afforded a triyne intermediate **IN-5** first. Subsequent intramolecular carbopalladations would afford the final product. We reasoned that CuI may play a role in the intramolecular insertion of **IN-1** forming **IN-2**.

In order to unveil the mechanism of this reaction, some control experiments have been carried out (Scheme 4). The reaction of **4a** with terminal propargyl alcohol **5d** afforded a tricyclic lactone product **7ad** in 8% yield, together with a benzofuran product **8ad** in 49% yield. Obviously, the product **7ad** was formed after the lactonization of the regioselective isomer of **8ad**. The reaction of **4b** with **5d** gave similar results. Both of the two reactions indicated that the **intermolecular process** should be the major mechanistic route, otherwise the tricyclic product **7** should be the main product. The reaction in the absence of CuI was complicated-the formation of the product was not detected after one hour under standard conditions (eq. c in Scheme 4), indicating the importance of CuI for this transformation. In addition, under the optimized conditions, when the reaction time was shortened from 3.5 h to one hour, bicyclic intermediate product **9ad** was indeed isolated (eq. d in Scheme 4). Moreover, when **7ad** was submitted to the standard conditions, the formation of **9ad** was not observed and 100% of **7ad** was recovered (eq. e in Scheme 4), ruling out the possibility of hydrolysis of lactone **7ad** forming **9ad**. These results further support the **intermolecular process** shown in Scheme 3.

Some synthetic applications have been performed to show the potential of the products (Scheme 5). The $Fe(NO_3)_3 \cdot 9H_2O$ catalyzed aerobic oxidation of **6ab** proceeded smoothly to give the aryl aldehyde **10aa** in 76% yield.[13] The tetra-cyclic lactone **11aa** was easily

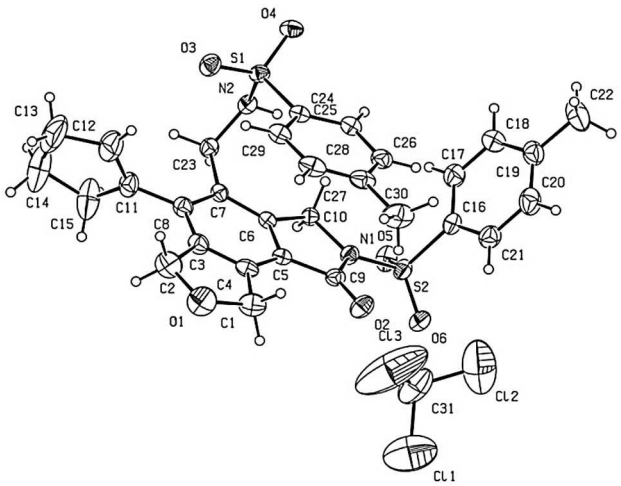
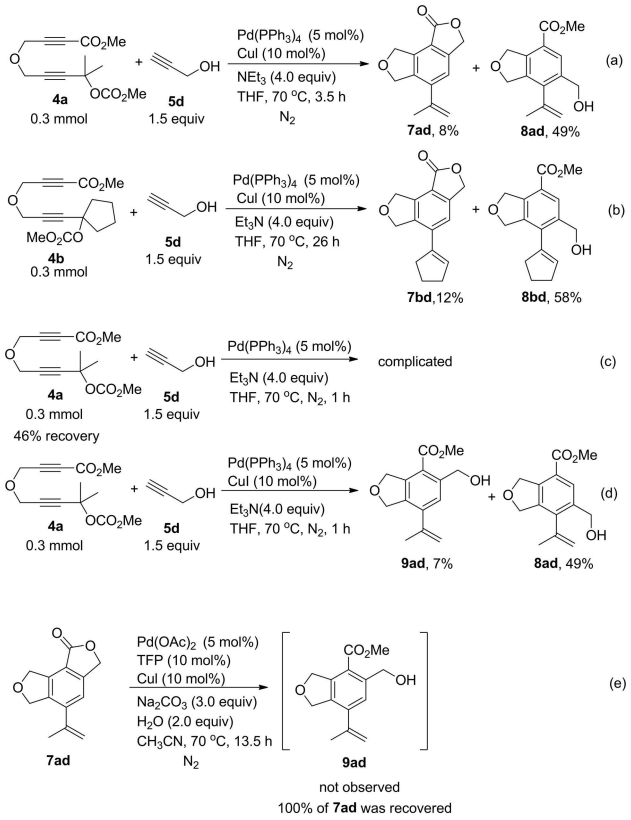
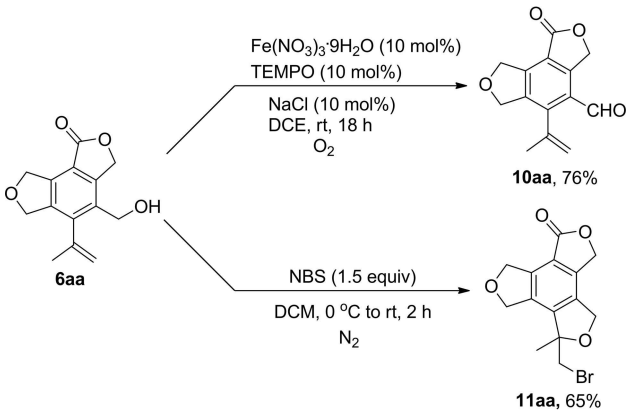


Figure 1. ORTEP representation of **6bb**·$CHCl_3$.



Scheme 4. Control experiments.



Scheme 5. Synthetic applications.

obtained through an electrophilic cyclization with NBS (1.5 equiv.).[14]

In summary, we have developed an efficient and stereoselective method for the construction of tricyclic lactones or lactams via a palladium-catalyzed 8-methyoxycarbonyl substituted 2,7-alkadiynylic carbonates with 2-butyne-1,4-diol or 2-butyne-1,4-disulfonamides. After the studies of control experiments, we reasoned that the reaction should preferentially go through intermolecular carbopalladations before the

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eventual lactonization. The useful functional groups such as alcohol, sulfonamide, and alkene in the products provided opportunities for the further transformation to more complicated polycycles. Further studies in this area are being pursued in our laboratory.

Experimental Section

General Experimental Methods

^{1}H NMR and ^{13}C NMR spectra were recorded in $CDCl_3$ with a Bruker AM 300 MHz NMR spectrometer (^{1}H at 300 MHz, ^{13}C at 75 MHz) using TMS (^{1}H, $\delta = 0$) or residual $CHCl_3$ (7.26 ppm) as the internal standards. IR spectra were recorded with a Perkin-Elmer 983G instrument. Elemental analyses were measured with a Carlo-Erba EA1110 elementary analysis instrument. Mass spectrometry was performed with an HP 5989 A system. High-resolution mass spectrometry was determined with a Finnigan MAT 8430 or Bruker APEXIII instrument. $Pd(PPh_3)_4$ was prepared according to the known literature.[1] $Pd(OAc)_2$ and CuI were purchased from *Adamas*. TFP was purchased from *J&K*. THF was distilled with Na wire using benzophenone as the indicator under N_2 before use. CH_3CN was distilled with CaH_2 under N_2 before use. All the temperatures are referred to the oil baths used.

Typical Procedure for 4-(hydroxymethyl)-5-(prop-1-en-2-yl)-6,8-dihydrobenzo[1,2-c:3,4-c′]difuran-1-(3*H*)-one 6 aa

Synthesis of **6 aa**: To a flame-dried Schlenk tube were added CuI (0.0193 g, 0.1 mmol), $Pd(PPh_3)_4$ (0.0577 g, 0.05 mmol), **5 a** (0.1041 g, 1.2 mmol), **4 a** (0.2671 g, 1 mmol)/THF (10 mL), and Et_3N (0.55 mL, d = 0.73 g/mL, 0.40 g, 4 mmol) sequentially under N_2 atmosphere. The reaction mixture was then stirred at 70 °C for 5 hours as monitored by TLC. The resulting mixture was filtrated through a short column of silica gel (2 cm) and eluted with ethyl acetate (20 mL × 3). After evaporation of the solvent, the residue was purified by chromatography on silica gel to afford **6 aa** (0.1673 g, 68%) (eluent: petroleum ether/ethyl acetate = 1/1 (500 mL)): solid; m.p. 197.6–200.1 °C (DCM/hexane); ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.53 (s, 2 H, OCH_2), 5.38 (s, 2 H, OCH_2), 5.36–5.32 (m, 1 H, one proton of $=CH_2$), 5.04 (s, 2 H, OCH_2), 4.96–4.88 (m, 1 H, one proton of $=CH_2$), 4.83 (s, 2 H, OCH_2), 2.42 (brs, 1 H, OH), 2.02 (t, J = 1.2 Hz, 3 H, Me); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.8, 146.2, 141.9, 141.3, 139.4, 136.8, 131.8, 118.8, 116.7, 72.6, 72.4, 71.2, 60.0, 23.2; IR (KBr) ν (cm^{-1}) 3428, 2953, 2922, 2867, 1751, 1455, 1421, 1381, 1347, 1272, 1124, 1086, 1050, 1009; MS (EI): m/z (%) 246 (M^+, 100), 231 [$(M-CH_3)^+$, 8.04]; Anal. Calcd. for $C_{14}H_{14}O_4$ (%): C 68.28, H 5.73; Found: C 67.88, H 5.85.

Acknowledgements

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[11] Crystal data for compound **6bb·CHCl$_3$**: $C_{30}H_{30}N_2O_6S_2 \cdot CHCl_3$; MW = 698.05; triclinic, space group P −1; final R indices [I > 2\s(I)], R1 = 0.0729, wR2 = 0.2048, R indices (all data) R1 = 0.0959, wR2 = 0.2307; a = 6.8943(4) Å, b = 15.7166(9) Å, c = 16.6797(8) Å; α = 110.045(5), β = 95.577(4), γ = 98.666(5); V = 1656.77(17) Å^3; T = 293 K; Z = 2; reflections collected/unique 10832/ 6054 (R_{int} = 0.0292); number of observations [> 2\s(I)] 4502, parameters: 399. CCDC-1881002 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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